PART 1 OF 2 — THE WHY

Historic Scripts to Modern Vision

Understanding the ModiTrans VLM and its datasets — a field guide for experienced software engineers.

Model: **historyHulk/ModiTrans-12B-Gemma-Teacher**

Datasets: **MoDeTrans** (real) · **SynthMoDe** (synthetic)

Paper: Kausadikar, Kale, Susladkar, Mittal — ICDAR 2025 · arXiv 2503.13060

ORIENTATION

Four artifacts, one project

THE PAPER

“Historic Scripts to Modern Vision” — peer-reviewed at **ICDAR 2025**. It defines the problem, the datasets, and a model framework called **MoScNet**.

THE MODEL

ModiTrans-12B-Gemma-Teacher on Hugging Face — a LoRA adapter over **google/gemma-3-12b-it**. This is the *teacher* from the paper's recipe.

MODETRANS (REAL)

2,043 real, scanned Modi-script document images, each paired with an expert-written Devanagari transliteration.

SYNTHMODE (SYNTHETIC)

2,043 machine-generated Modi images (2 fonts each) built from MoDeTrans' Devanagari text. The 'easy' twin of the real set.

60-SECOND SUMMARY

TL;DR — what this is

- **Goal:** turn a photo of handwritten **Modi script** (medieval Marathi) directly into modern **Devanagari** text — end-to-end, not character-by-character.
- **Why it's novel:** first system to do *direct full-document transliteration*; prior work mostly recognized single characters.
- **How:** a Vision-Language Model + **Knowledge Distillation** — a big 'teacher' LLM trains a tiny 'student' that is **163× smaller** yet scores higher.
- **Result:** best student (MoScNet-XL, 429M params) reaches **BLEU 51.0** on the real set — a strong research baseline, not a solved problem.
- **Availability:** datasets + the **teacher** adapter are public (MIT, gated by a click-through). The distilled student weights are *not* on the Hub.

HOW TO READ THIS DECK

The path we'll take

- **The WHY** — the problem, the data, the model decisions, and an honest assessment.
- **The reviewer's lens** — the sharp questions you should ask before trusting any model/dataset.
- **Using it** — what is actually released and how to reach it.

Part 2 (separate deck) is The HOW — every concept here returns there with runnable code: data pipeline, LoRA + distillation training, evaluation, and a from-scratch playbook.

01

The WHY

What problem is this solving, and why should anyone build it?

THE SUBJECT

What is Modi script?

- The official script for writing **Marathi** during the Maratha era; used across Maharashtra until India's pre-independence period.
- Six historical phases — the model uses the three with surviving documents: **Shivakalin** (17th c.), **Peshwekalin** (18th–19th c.), **Anglakalin** (1818–1952).
- Superseded by **Devanagari** — the script Marathi & Hindi use today.

> ⓘ The uploaded figure shows the exact task: a Modi manuscript (left) becoming readable Devanagari (right).

Modi (Modi script) → Marathi in Devanagari. Source: user-provided figure.

THE MOTIVATION

Why this matters — the scale

~40M	few	83M
DOCUMENTS UN-TRANSLITERATED	LIVING EXPERTS WHO CAN READ IT	MARATHI SPEAKERS (2011 CENSUS)

- Modi documents hold **land & property records, Ayurvedic medicine, administrative history** — primary evidence of medieval India.
- Many are **fragile and decaying**; manual transliteration can't scale to 40 million documents with a handful of experts.
- An automated bridge enables **digitization, search, and historical research** — and helps revive cultural literacy.

THE DIFFICULTY

Why Modi is hard for machines

PROPERTIES OF THE SCRIPT

- **Shirorekha:** a top line spanning the page; letters hang from it.
- **No word spaces:** cursive, continuous writing — no clean word boundaries.
- **Ligatures & angular strokes** — many shapes blur together.
- **Look-alike characters** and almost no punctuation.

PLUS REAL-WORLD DECAY

- Faded ink, torn pages, water/age damage.
- Many writing styles across eras (Chitnisi, Mahadevpanti, Bilavalkari, Ranadi).
- Tiny, scattered, hard-to-access source documents.

! These are exactly why a **robust** model is hard — and why a synthetic 'clean' dataset alone is not enough.

CONCEPT CHECK

OCR vs Transliteration vs Translation

TASK	WHAT IT DOES	EXAMPLE
OCR	Image of text → the *same* text as characters	Photo of 'मौजे' → “मौजे”
Transliteration	Map symbols of one *script* to another, *sound preserved*	Modi glyphs → Devanagari “मौजे”
Translation	Change the *language/meaning*	Marathi → English “the village”

This project is **transliteration**: Modi-script image → Devanagari text. The *language stays Marathi* — only the script changes. (The model also happens to be good at generic OCR.)

THE GAP

What was missing before this work

- Prior Modi research mostly did **single-character recognition** on character datasets (MODI-HChar, Barakhadi, etc.).
- One prior effort used an LSTM on **synthetic-only** data — which misses real handwriting complexity.
- **No public dataset** paired real Modi *documents* with full Devanagari transliterations.

✓ **The thesis:** jump from 'recognize a letter' to 'read a whole document.' That leap is the contribution — and it needs a new dataset to even measure it.

THE IDEA

The whole project in one sentence

Treat reading Modi as an **image-to-text** problem and solve it with a **Vision-Language Model**, then **distill** that ability into a tiny model cheap enough to deploy.

INPUT

A preprocessed image strip of Modi script (a few lines).

MODEL

Vision encoder → projector → language model (teacher), distilled into a small student.

OUTPUT

A Devanagari text string — the transliteration.

02

The datasets

Where the knowledge comes from — and the catch you must notice.

TWO DATASETS

MoDeTrans vs SynthMoDe at a glance

ASPECT	MODETRANS	SYNTHMODE
Nature	**Real** scanned manuscripts	**Synthetic** font-rendered images
Rows	2,043	2,043 (mirrors MoDeTrans text)
Columns	filename · image · text	filename · image1 · image2 · text
Image source	Archives, experts, MODI-HHDoc	Rendered from Devanagari text via 2 Modi fonts
Labels	Hand-written by Modi experts	Reused from MoDeTrans (same strings)
Role	The real benchmark (hard)	Augmentation / sanity twin (easy)
License	MIT	MIT

Inside the real dataset

SCHEMA (PER ROW)

- **filename** — e.g. `1.jpg`
- **image** — a denoised Modi strip (~3–4 lines, ~40 words)
- **text** — the Devanagari transliteration (10–452 chars)

ERAS COVERED

- Shivakalin — 891 imgs (43.6%)
- Peshwekalin — 944 imgs (46.2%)
- Anglakalin — 208 imgs (10.2%)

Adyakalin & Yadavkalin eras omitted — too few well-preserved documents survive.

Provenance — where the 2,043 images came from

924
FROM MODI-HHDOC DATASET

903
DONATED BY MODI EXPERTS

216
SOURCED FROM ARCHIVES

i Documents live in places like the **Bharat Itihas Sanshodhak Mandal** (Pune), the Institute of Oriental Studies, and state archives. Access scarcity is a core reason this dataset is valuable.

✓ For the 924 MODI-HHDoc images (images only), the team **generated all transliterations manually** — the labels are original work.

A real example (authentic label)

FILENAME	TEXT (DEVANAGARI)
1.jpg	स्वासनिधी राजमान्य राजश्री माधवराव पंडित प्रधान तां मोकदम व रयान मौजे वाठार पाा फलटण सुा इसने सबैन मया व अलफ तु

This is verbatim from the dataset viewer. The matching **image** is a scanned Modi strip; the text is what an expert read from it. The same string appears as row 1.jpg in **both** datasets — remember that.

How the labels were made — human-in-the-loop

Modi document image → **Expert transliterates (to Devanagari)** → **Cross-verify 2nd expert pass** → **Review & correct** → **Final pair**

- Transliteration done by experienced Modi experts (the paper credits **Mrs. Kanchan Karai**).
- **Clear guidelines** to keep it faithful and bias-free; a **cross-verification** pass catches errors.
- A team of **10 annotators** manually checked the train/test/val split for overlap.

✓ This is a textbook **HITL labeling pipeline**: expert produces → second expert verifies → reviewers correct. Slow, costly, high-trust.

Image preprocessing pipeline (real docs)

Doc image → Split into 3–4 strips → Grayscale → Denoise bilateral k=3 → Manual fix Photoshop, if needed

↳ Adaptive (Gaussian) thresholding → Deskew → Clean strip → dataset

i Gaussian adaptive threshold uses **$t = k \cdot \sigma$** ($0.5 < k < 2$). The 'manual fix' branch (human + Photoshop) is the honest, un-glamorous reality of historical-document ML.

SYNTHMODE

What it is and why it exists

- A **synthetic** twin: Modi images generated by a computer, not scanned.
- Built because a prior synthetic Modi set (Kundaikar et al.) was **never released** — so the authors recreated one.
- Its real job: **prove a point** — synthetic Modi is far easier than real Modi, which justifies why MoDeTrans is needed.

SCHEMA (PER ROW)

- **filename** — `1.jpg` (mirrors MoDeTrans)
- **image1** — MarathiCursive font render
- **image2** — Noto-Sans-Modi font render
- **text** — same Devanagari string as MoDeTrans

SYNTHMODE

How it was generated — programmatically

Devanagari text from MoDeTrans → Render with MarathiCursive font → image1

Same text → Render with Noto-Sans-Modi font → image2

! Key point: the pipeline goes **text → image**. So the 'ground-truth' for a synthetic image is just the text you started from — no human reading required. That makes it cheap, clean, and **too easy** to be a real test.

Part 2 reproduces this exact engine in ~30 lines of Pillow code.

REAL VS SYNTHETIC

The insight the comparison buys you

SYNTHETIC (SYNTHMODE)

- Uniform fonts, clean strokes, no decay.
- Models score **higher** here (BLEU 53 vs 51).
- Good for pre-training & sanity checks.

REAL (MODETRANS)

- Diverse handwriting, noise, faded ink.
- The **hard benchmark** that actually matters.
- Where the research value lives.

✓ Lesson for any ML project: **synthetic data flatters your model.** Always keep a real, messy held-out set as the score that counts.

CRITICAL LENS

⚠ The data-leakage subtlety you must notice

- SynthMoDe's **text labels are copied from MoDeTrans** — identical strings, same filenames.
- So a model trained on MoDeTrans has **already seen every target sentence** that appears in SynthMoDe.
- The paper's 'zero-shot on SynthMoDe' means **new images of text the model already learned** — not unseen *content*.

⚠ This isn't fraud — it's a legitimate cross-render generalization test, and the authors took strong steps against *split* leakage (10 annotators). But as a reviewer you must state it precisely: **image distribution differs; text targets overlap.** A truly clean test needs held-out *sentences*, not just held-out images.

03

The model

From a generic Gemma to a Modi reader — the tree, the parts, the distillation.

MODEL TREE

What the Hugging Face page tells you

google/gemma-3-12b-pt pretrained base	→	**google/gemma-3-12b-it** instruction-tuned	→	**ModiTrans-12B-Gemma-Teacher** LoRA / PEFT adapter

BASE MODEL

The foundation everything builds on. **gemma-3-12b-pt** is Google's raw pretrained 12B multimodal model.

INSTRUCTION-TUNED

-it follows prompts/chat. The team fine-tunes *this* one.

THE ADAPTER

Tiny LoRA weights that teach Gemma the Modi→Devanagari task. This is the released artifact.

CONCEPT

What is a 'base model' and why start there?

- A **base model** is a large network already trained on enormous data — it knows language, vision, and (crucially) **Devanagari**.
- You **don't train from scratch**: you adapt. This is *transfer learning*.
- Why Gemma-3-12B-it: multimodal (sees images), open weights, and its tokenizer already covers Hindi/Devanagari — so Marathi output 'just works' without a new vocabulary.

✓ Reusing the tokenizer is a real design decision: Marathi & Hindi share Devanagari, so the model inherits subword knowledge instead of relearning the script. Less complexity, better results.

IMPORTANT CAVEAT

⚠ Paper recipe ≠ released file

WHAT THE PAPER EXPERIMENTS USE

- Teachers: **LlaMA-2 / LlaMA-3 / Mistral / Phi-3**.
- Best teacher: **LlaMA-3 70B** (BLEU 49.4).
- Student: custom **MoScNet-XL**, 429M params (BLEU 51.0).

WHAT'S ON HUGGING FACE

- A **Gemma-3-12B-it** teacher LoRA adapter.
- The **distilled student is NOT published**.
- So the famous '163× smaller' model isn't downloadable here.

⚠ Don't conflate the two. The released checkpoint is a **Gemma teacher** — useful and runnable, but a different base than the paper's headline numbers. Always reconcile 'paper claims' with 'actual files'.

CONCEPT

What is a VLM (Vision-Language Model)?

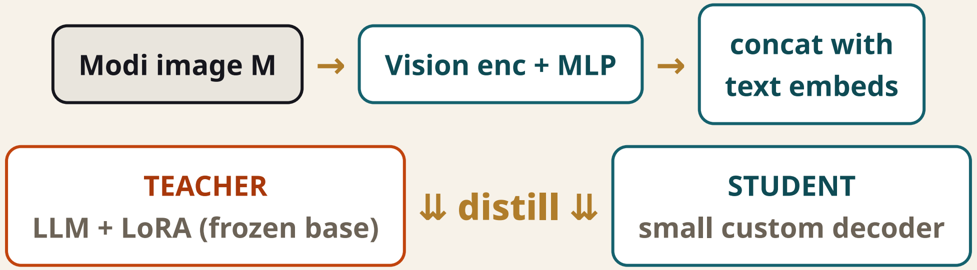
Image → Vision encoder SigLIP → Projector MLP → Language model LLM decoder → Text out

- **Vision encoder** turns pixels into image tokens I.
- **Projector (MLP)** maps image tokens into the LLM's embedding space → Z_I.
- Those are concatenated with text embeddings Z_L and fed to the LLM, which generates Devanagari.

Ablation result: **SigLIP** beat CLIP, ViT-L, and Swin as the vision encoder — its image tokens align best with text.

ARCHITECTURE

MoScNet — the teacher/student picture



The teacher is a big LLM fine-tuned with LoRA. The student is a small decoder that learns to **mimic the teacher** — both supervised by the true Devanagari text. At deploy time you ship only the student.

CONCEPT

Knowledge Distillation — the headline trick

- A **big teacher** is accurate but expensive. A **small student** is cheap but weak — unless it learns from the teacher.
- The student is trained to match (a) the true text *and* (b) the teacher's internal signals.
- Outcome: **MoScNet-XL (429M)** beats its **70B** teacher on BLEU while being **163× smaller**.

163×

FEWER PARAMETERS THAN TEACHER

51.0

STUDENT BLEU > 49.4 TEACHER

✓ Why a student can beat its teacher: distillation transfers 'dark knowledge' (soft signals), and a task-specific small model can specialize without the teacher's general-purpose baggage.

CONCEPT

LoRA / PEFT — fine-tuning without the bill

- Full fine-tuning of a 12B–70B model updates **billions** of weights — huge memory & compute.
- **LoRA** freezes the base and injects small trainable low-rank matrices into each layer (here, **rank 64**).
- You train <1% of parameters, swap adapters per task, and add **zero** extra inference latency once merged.

WHY IT FITS HERE

The released file *is* a LoRA adapter — a few MB that sit on top of frozen Gemma. That's why you load **base + adapter**, never a full new model.

THE NOVELTY

What's new in the student model

PARALLEL ATTENTION

Attention and MLP run **side-by-side** in each block (not sequentially). Removing it drops BLEU 51.0 → 49.2.

QK-NORMALIZATION

Normalize attention **queries & keys** to stabilize training. Removing it drops BLEU 51.0 → 50.1.

i The paper claims this is the first use of parallel-attention + QK-norm in a **causal** (text-generating) setting. You don't need to memorize the math — just know *which* choices the ablations proved earned their place.

DESIGN RATIONALE

Why these specific building blocks?

CHOICE	WHY
Gemma-3-12B-it (released) / LlaMA-3 (paper)	Open, strong, already trained on Devanagari/Hindi → reuse tokenizer, no new vocab
SigLIP vision encoder	Best cross-modal alignment in ablations vs CLIP/ViT-L/Swin
Knowledge Distillation	Transliteration doesn't need a 70B model at inference — shrink it
LoRA (rank 64)	Fine-tune a giant model cheaply; ship a tiny adapter
Parallel attention + QK-norm	Squeeze more accuracy from a small student (proven by ablation)

04

Training & results

The knobs they turned, the numbers they got, and how to read them honestly.

TRAINING SETUP

The configuration that produced the model

SETTING	VALUE
Hardware	16 nodes × 8 NVIDIA H100 = 128 GPUs, BF16
Optimizer / schedule	AdamW + cosine learning-rate decay
Learning rate	Teacher 1e-6 · Student 1e-4
Batch	4 per GPU × gradient-accumulation 8
LoRA rank	64 (teacher adapter)
Input image size	128 × 256 (paper) — note: model card example resizes to 1024×512
Loss (student)	Cross-Entropy + L2 + KL-divergence

! Spot the discrepancy to investigate: the paper trains at **128×256**, but the HF inference snippet resizes to **1024×512**. Always match preprocessing to how the checkpoint was trained.

THE LOSS

What the model is actually optimizing

TEACHER LOSS

Cross-Entropy only — predict the next Devanagari token correctly. Backprops into the LoRA matrices.

STUDENT LOSS = CE + L2 + KL

- **CE** — match the true text (the main job).
- **L2** — pull student's hidden states toward the teacher's.
- **KL** — match the teacher's output *distribution*, not just its top pick.

✓ Ablation proof: CE-only = 48.0 BLEU; +L2 = 50.4; +KL = 50.3; all three together = **51.0**. Each distillation signal adds something.

RESULTS

Transliteration quality (BLEU ↑)

METHOD	MODETRANS (REAL)	SYNTHMODE (SYNTH)
CRNN (CTC baseline)	27.3	28.7
ViTSTR	31.2	33.4
LlaMA-2 13B + SigLIP	40.2	42.1
LlaMA-3 8B + SigLIP	43.1	46.2
LlaMA-3 70B + SigLIP	49.4	52.5
MoScNet-XL (KD, LlaMA-3 70B teacher)	**51.0**	**54.0**

i BLEU is an n-gram overlap score (0–100). Higher is better, but ~51 is a **strong research result, not human parity**. Synthetic always scores higher — the easy-data effect again.

RESULTS

Also strong at generic OCR (char-accuracy ↑)

METHOD	ICDAR13	ICDAR15
CRNN	90.9	70.8
ABINet	95.0	79.0
ParSeq-A	96.2	82.2
CoFormer-A	96.3	82.9
MoScNet-XL	**97.9**	**85.5**

✓ Trained on MJSynth, evaluated on standard scene-text benchmarks. Evidence the architecture generalizes beyond Modi — a nice robustness signal for the design.

RESULTS

What the ablations actually proved

VISION ENCODER

CLIP 49.3 · ViT-L 50.1 · Swin 50.7 · **SigLIP 51.0**

STUDENT SIZE

S 43.3 · M 48.4 · L 50.0 · **XL 51.0** (44M→429M)

LOSS FUNCTION

CE 48.0 · +L2 50.4 · +KL 50.3 · **all 51.0**

ARCHITECTURE

no parallel-attn 49.2 · no QK-norm 50.1 · **full 51.0**

✓ Every headline component is justified by a controlled experiment. This is good practice — when *you* review or build, demand the same: change one thing, measure the delta.

HONEST READING

How good is 'BLEU 51' really?

- It clears all prior baselines — a genuine state-of-the-art for this task.
- But BLEU ~51 means **frequent character-level errors remain**; the paper itself lists confusions (bha/ma, ka/pha, missing anusvāra dots).
- It is an **assistive** tool: great for a first pass that a human expert then corrects — not an autonomous archivist.

! Map this onto your own domain instincts: a model that's 'best-in-class' can still be wrong often enough that **human-in-the-loop** is mandatory. Set expectations accordingly.

The cost & footprint nobody shows you

128
H100 GPUS

~40 kg
CO_2 PER TRAINING HOUR

~52 kg
CO_2 FOR THE RUN (≈1.3H)

- The authors estimate and **report** CO_2 — a maturity signal worth imitating.
- Distillation isn't just about speed: a 429M model is dramatically cheaper & greener to **run** than a 70B one.

When you propose a model, budget compute and carbon up front. 'It works' and 'we can afford to run it' are two different approvals.

05

Teach, learn, assess

What to take away as a teacher, a student, and a skeptical reviewer.

AS A TEACHER

What this project is perfect for teaching

CONCEPTS

- Transfer learning & base models
- VLMs: vision encoder + projector + LLM
- Knowledge distillation (teacher→student)
- LoRA / PEFT efficient fine-tuning

PRACTICES

- Real vs synthetic data trade-offs
- Human-in-the-loop labeling & QA
- Ablation-driven design decisions
- Honest evaluation & its limits (BLEU)

AS A STUDENT

Your skills ladder for this work

- **Rung 1 — Use it:** load base + LoRA, run inference on an image (Part 2 §1).
- **Rung 2 — Reproduce data:** implement the OpenCV preprocessing & the synthetic font-render engine.
- **Rung 3 — Fine-tune:** attach LoRA to a VLM and train on image→text pairs.
- **Rung 4 — Distill:** stand up a teacher→student loop with CE + L2 + KL.
- **Rung 5 — Evaluate & harden:** BLEU/CER, leakage checks, robustness & privacy probes.

✓ Each rung maps to a section of Part 2 with code. Climb in order — don't start at distillation.

INDEPENDENT ASSESSMENT

Strengths — what's genuinely well done

- ✓ **Fills a real gap:** first real paired dataset + first direct document transliteration for Modi.
- ✓ **Rigorous data work:** expert labeling, cross-verification, manual split auditing against leakage.
- ✓ **Efficiency story:** distillation to a 429M model is practical and deployable.
- ✓ **Disciplined evaluation:** baselines across RNN→LLM, four ablation studies, even a CO_2 estimate.
- ✓ **Open:** datasets + teacher adapter public under MIT; reproducible recipe in the paper.

INDEPENDENT ASSESSMENT

Limitations & risks — what to watch

- **Released ≠ paper:** only a Gemma *teacher* adapter is on the Hub; the star student isn't published.
- **Metric gap:** BLEU only — no **CER/WER**, and no robustness numbers on damaged manuscripts.
- **Text-target overlap** between MoDeTrans & SynthMoDe muddies 'zero-shot' claims.
- **Small expert pool** for ground truth → potential systematic bias in 'correct' readings.
- **Scale:** 2,043 documents is small for a 12B-class VLM; coverage skips two eras.
- **Preprocessing discrepancy** (128×256 vs 1024×512) can quietly hurt your inference.

IF STARTING NEWLY

How I'd approach this from zero

- **Define the real test first:** hold out whole *documents/sentences*, never just images.
- **Invest in the synthetic engine** (more fonts, realistic noise) to pre-train cheaply before touching scarce real data.
- **Pick a small modern open VLM** and LoRA-tune it before chasing distillation.
- **Add CER/WER** from day one; BLEU alone hides character mistakes.
- **Bake in HITL:** ship a correction UI so experts fix outputs and feed corrections back.
- **Measure robustness & privacy explicitly** — don't assume, test.

i Full code for each of these is in Part 2's 4-phase playbook.

REVIEWER'S CHECKLIST

The questions you should ask — and the answers here

QUESTION TO ASK	WHAT THE EVIDENCE SHOWS
Data leakage — overlap between SynthMoDe & MoDeTrans?	Yes at the **text-label level** (synthetic is rendered from the real text). Image distributions differ; splits were manually de-duplicated.
Zero-shot on unseen scripts (e.g. Brahmi)?	**Not evaluated.** 'Zero-shot' in the paper = same text, synthetic images. No cross-script generalization is claimed or tested.
Expert bias — how were labels validated?	Expert transliteration + second-expert cross-verification + 10-annotator split review. But the **expert pool is small**; no inter-annotator agreement score is reported.
Robustness — CER on water/fire-damaged docs?	**Not reported.** Only BLEU (and scene-text char-accuracy). No damage-stratified error analysis exists.
Privacy — does it memorize land-record data?	**Not tested.** Authors state no sensitive/personal data was used; records are centuries old. Memorization was not probed.

REVIEWER'S MINDSET

Why those five questions generalize

- **Leakage** → 'is my test truly unseen?' applies to *every* ML project.
- **Zero-shot** → 'what did it actually generalize to vs memorize?'
- **Label validation** → 'who decided ground truth, and could they be wrong together?'
- **Robustness** → 'how does it fail on the ugly inputs I'll really see?'
- **Privacy** → 'can the model leak its training data?'
- Healthcare/finance parallel: same five questions decide whether a model is **safe to deploy**.

✓ A good reviewer doesn't ask 'is the BLEU high?' — they ask 'what's **not** measured, and what breaks in production?'

06

Using it

Is it public, what do you actually get, and how do you run it?

AVAILABILITY

Is it open? Mostly yes — with footnotes

PUBLIC

- License: **MIT** (model + both datasets).
- Datasets have live web viewers — browse rows now.
- Inference + training recipe in the paper.

FOOTNOTES

- Model is **gated**: click-through to accept terms + HF login.
- You get a **LoRA adapter**, not a standalone model — load it on Gemma-3-12B-it.
- The distilled **student is not released**.

USING IT

The minimal path (full code in Part 2)

load_modi_teacher.py — base + adapter, the only way it works

```
from transformers import AutoModelForImageTextToText, AutoProcessor
from peft import PeftModel
from PIL import Image
import torch

base = "google/gemma-3-12b-it"
adapter = "historyHulk/ModiTrans-12B-Gemma-Teacher"

model = AutoModelForImageTextToText.from_pretrained(
    base, torch_dtype=torch.bfloat16, device_map="cuda:0")
model = PeftModel.from_pretrained(model, adapter)      # attach the LoRA
processor = AutoProcessor.from_pretrained(base)

img = Image.open("modi_strip.jpg").convert("RGB").resize((1024, 512))
# build a chat message with image + instruction, then model.generate(...)
```

i You need a GPU with enough memory for a 12B model in bf16 (~24GB+), Hugging Face login, and an accepted license. Part 2 covers the full decode loop and the OOM/dtype gotchas.

EXPECTATIONS

What you get vs what you might assume

YOU MIGHT ASSUME...	REALITY
A ready-to-run small fast model	A **LoRA on a 12B VLM** — big and slow without the student
The 'BLEU 51 / 163× smaller' model	That's the **unreleased student**; this is the **teacher**
A finished product	A **research checkpoint** — assistive, error-prone, needs HITL
Plug-and-play API	Self-host only; no hosted inference provider yet

07

Wrap-up

Takeaways, glossary, and where to go next.

TAKEAWAYS

The ten things to remember

- Modi→Devanagari = **transliteration**, framed as image→text.
- **MoDeTrans** (real, hard) is the benchmark; **SynthMoDe** (synthetic, easy) supports it.
- Synthetic is rendered **from** the real text → watch the leakage line.
- Model = **VLM + Knowledge Distillation + LoRA**.
- Student beats teacher: **163× smaller, BLEU 51**.
- Released artifact = **Gemma teacher adapter**, not the student.
- BLEU only → **no CER, no damage robustness** numbers.
- Strong data hygiene, but **small expert pool & 2k docs**.
- It's **assistive**: human-in-the-loop required.
- Public & MIT, but **gated** and self-hosted.

GLOSSARY

Plain-English definitions

VLM	Vision-Language Model — sees images, writes text	BLEU	0–100 text-overlap score; higher = closer to reference
Base model	Big pretrained net you adapt, not train from scratch	CER/WER	Character/Word Error Rate — what's missing here
LoRA / PEFT	Cheap fine-tuning: train tiny add-on matrices	Shirorekha	The top line Modi letters hang from
KD	Knowledge Distillation — student learns from teacher	Anusvāra	The dot diacritic the model sometimes drops
SigLIP	The vision encoder that turns pixels into tokens	Adapter	The released LoRA weights placed on Gemma

SOURCES

Go to the primary materials

- **Paper:** arXiv 2503.13060 · Springer ICDAR 2025 (doi:10.1007/978-3-032-04630-7_3)
- **Model:** huggingface.co/historyHulk/ModiTrans-12B-Gemma-Teacher
- **Real data:** huggingface.co/datasets/historyHulk/MoDeTrans
- **Synthetic data:** huggingface.co/datasets/historyHulk/SynthMoDe
- **Fonts used:** Noto Sans Modi · MarathiCursive (MihailJP)

i Authors: Harshal Kausadikar & Tanvi Kale (co-first, independent researchers), Onkar Susladkar (Yellow.AI), Sparsh Mittal (IIT Roorkee). Modi expertise: Mrs. Kanchan Karai.

NEXT

Part 2 — The HOW awaits

You now understand **why** this exists and **what** to trust. Part 2 turns every concept here into **runnable code**:

- Inference with the released teacher — end to end.
- The data pipeline & synthetic engine, reproduced.
- LoRA fine-tuning + the full distillation loop.
- Evaluation, leakage/robustness/privacy probes.
- A 4-phase build-from-scratch playbook.